No announcements this week

Good luck on lecture quiz 3

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1): min/max $f = 9xe^y$ if $2 = x^2+y^2$ using Lagrange

We have $g := x^2+y^2 = 2$ fixed & want to min/max f. Now we apply Lagrange multipliers:

$$\nabla f = 9\langle e^y, xe^y\rangle = \lambda \nabla g = 2\lambda\langle x, y\rangle$$

$$\Rightarrow \left\{ \begin{array}{l} 9e^y = 2\lambda x \\ 9xe^y = 2\lambda y \end{array} \right\}.$$

Notice $9e^y$ appears twice. Let's sub in for one appearance:

$$2\lambda y = x(9e^y) = 2\lambda x^2 \Rightarrow \lambda y = \lambda x^2$$

$\Rightarrow y = x^2$ or $\lambda = 0$

$\lambda = 0$: $9e^y = 2\lambda x = 0 \Rightarrow e^y = 0$,

which has no solutions

$y = x^2$: We know $2 = x^2 + y^2 = x^2 + x^4$

$\Rightarrow 0 = x^4 + x^2 - 2 = (x^2 - 1)(x^2 + 2)$

$\Rightarrow x^2 = 1, -2$. But $x \in \mathbb{R}$, so $x^2 = 1$

$\Rightarrow x = \pm 1$, and in either case $y = 1$,

So we get $(\pm 1, 1)$ as candidates

Now we must check values of f:

$f(1,1) = 9 \cdot 1 \cdot e^1 = 9e$, $f(-1,1) = -9e$,

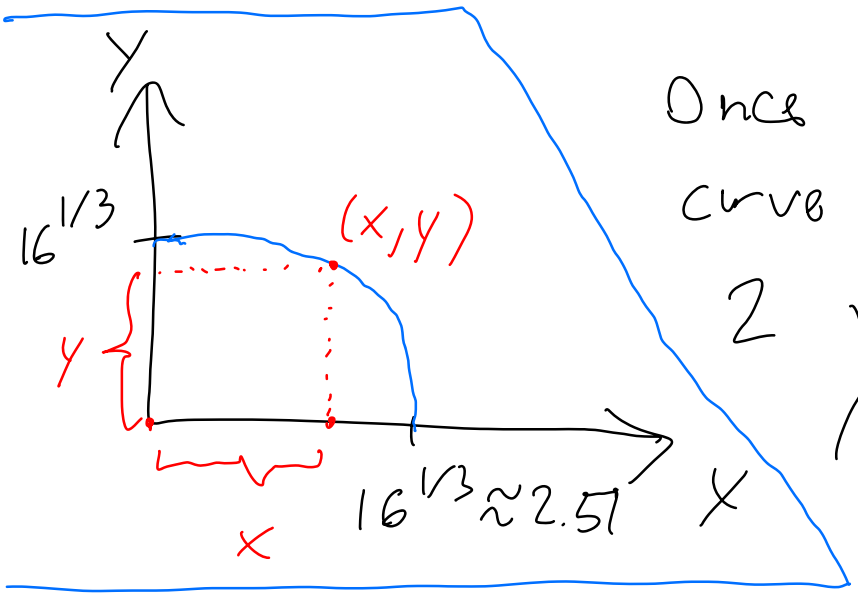
So min is $-9e$ & max is $9e$.

2: Rectangle w/ vertices at origin, on x^+ axis, on y^+ axis, and on curve $3x^2 + y^2 = 6$ in 1st quadrant. Use Lagr

$x^3 + y^3 = 16$ [illegible]

angle to find max area of rectangle.

Since this is a geometry problem, 1st step is to graph the region.



Once point (x,y) on curve is fixed, other 2 vertices must be $(0,y)$ & $(x,0)$, and the area is $f := xy$. We want to maximize f subject to $g := x^3 + y^3 = 16$, let's use Lagrange:

$$\nabla f = \langle y, x \rangle = \lambda \nabla g = 3\lambda \langle x^2, y^2 \rangle$$

$$\rightarrow \left\{ \begin{array}{l} y = 3\lambda x^2 \\ x = 3\lambda y^2 \end{array} \right\}$$, now let's sub 1st equation into 2nd:

$$x = 3\lambda y^2 = 3\lambda (3\lambda x^2)^2 = 27\lambda^3 x^4 \Rightarrow$$

$x=0$ or $27\lambda^3x^3=1 \Rightarrow x=\frac{1}{3\lambda}$

$x=0$: $y=3\lambda x^2=0 \Rightarrow$ area $=0$

$x=\frac{1}{3\lambda}$: $y=3\lambda x^2=\frac{3\lambda}{(3\lambda)^2}=\frac{1}{3\lambda}$,

now let's solve for λ. Plug into g:

$16=x^3+y^3=\frac{2}{(3\lambda)^3} \Rightarrow (3\lambda)^3=\frac{1}{8}=2^{-3}$

$\Rightarrow 3\lambda=\frac{1}{2} \Rightarrow \lambda=\frac{1}{6}$, so

$x=\frac{1}{3\lambda}=2$, $y=2$, area $=4$,

so maximum area is 4 for a 2×2 square.

3: min/max $f=2x^2-y^2+6y$ on the disk $x^2+y^2\le 16$

Just like last week, solve for critical points inside region & then check boundary.

$\vec{0}=\nabla f=\langle 4x, 6-2y\rangle \Rightarrow x=0,\ y=3,$

and $f(0,3) = 6\cdot 3 - 3^2 = 18-9 = 9$.

Now let's check boundary $x^2+y^2=16$, in which case $x=4\cos\theta$, $y=4\sin\theta$ parametrizes the circle. Then $f =$

$2x^2 - y^2 + 6y = 32\cos^2\theta - 16\sin^2\theta + 24\sin\theta =$

$-16 + 48\cos^2\theta + 24\sin\theta = -16 + 24(\sin\theta + 2 - 2\sin^2\theta)$.

$= -16 + 24(-2t^2+t+2)$, $t = \sin\theta$, so we want to $\min/\max$ $h = -2t^2+t+2$ on $[-1,1]$.

$0 = \frac{dh}{dt} = -4t+1 \Rightarrow t = \frac{1}{4}$, in which case $h(\frac{1}{4}) = -\frac{1}{8} + \frac{1}{4} + 2 = \frac{17}{8}$. Also, check the endpoints ± 1:

$h(1) = 3-2 = 1$, $h(-1) = -2+1 = -1$,

so $\min h = -1$, $\max h = \frac{17}{8}$. But

$f = -16 + 24h$, so $\min f = -40$ &

$\max f = -16 + 24\cdot\frac{17}{8} = 51-16 = 35$ on

boundary.

By comparing all values,

$$\min f = -40 \quad \& \quad \max f = 35$$

6: Average of $f = e^{2x-y}$ on rectangle $R = [0, \ln 3] \times [0, \ln 2]$

Recall formula: $\text{avg}_R f = \frac{1}{\text{area}(R)} \iint_R f \, dA$

$\text{area}(R) = \text{length} \cdot \text{width} = \ln 3 \cdot \ln 2$

$$\iint_R f \, dA = \int_0^{\ln 2} \int_0^{\ln 3} e^{2x-y} \, dx \, dy =$$

$$\int_0^{\ln 2} \frac{1}{2} e^{2x-y} \Big|_0^{\ln 3} dy = \frac{1}{2} \int_0^{\ln 2} \left(e^{2\ln 3 - y} - e^{-y}\right) dy$$

$$= \frac{1}{2} \int_0^{\ln 2} e^{-y} \left(e^{\ln 9} - 1\right) = 4 \int_0^{\ln 2} e^{-y} \, dy$$

$$-4 e^{-y} \Big|_0^{\ln 2} = -4\left(e^{-\ln 2} - 1\right) =$$

$= -4e^{-1} \Big|_0 = -4(e^{-1} - 1) =$

$-4\left(\frac{1}{2} - 1\right) = 2$, so $\boxed{\text{avg} = \frac{2}{(\ln 2)(\ln 3)}}$

For average value problems, it may help to do a common sense check for mistakes.

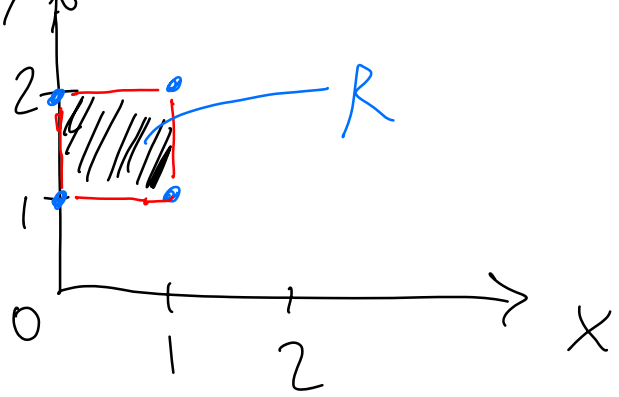
$2x - y \approx 2\left(\frac{1}{2}\ln 3\right) - \frac{1}{2}\ln 2 = \ln 3 - \ln\sqrt{2}$

$= \ln \frac{3}{\sqrt{2}}$, so $e^{2x-y} \approx \frac{3}{\sqrt{2}} \approx 2.12$,

so our answer is reasonable. But if you got an estimate that is far off your answer, then go back & check to see if you made any mistakes.

4c: $\iint_R \frac{dA}{(2x+3y)^2}$, R vertices (0,1), (1,1), (1,2), (0,2)

1st Step: find bounds on x&y



We can graph the rectangle by graphing vertices, and notice that $0 \le x \le 1$ and $1 \le y \le 2$

2nd Step: plug in xdy bounds, solve

$$\iint_R \frac{dA}{(2x+3y)^2} = \int_1^2 \int_0^1 \frac{1}{4} \frac{dx\,dy}{(x+1.5y)^2} =$$

$$\int_1^2 -\frac{1}{4(x+1.5y)}\Big|_{x=0}^{x=1} = \frac{1}{4}\int_1^2 \left(\frac{1}{1.5y} - \frac{1}{1.5y+1}\right) dy$$

$$= \frac{1}{6}\int_1^2 \left(\frac{1}{y} - \frac{1}{y+\frac{2}{3}}\right) dy = \frac{1}{6}\ln|y| - \ln\left|y+\frac{2}{3}\right|\Big|_1^2$$

$$= \frac{1}{6}\ln\left|\frac{y}{y+\frac{2}{3}}\right|\Big|_1^2 = \frac{1}{6}\left(\ln\frac{2}{8/3} - \ln\frac{1}{5/3}\right)$$

$$= \frac{1}{6}\left(\ln\frac{3}{4} - \ln\frac{3}{5}\right) = \boxed{\frac{1}{6}\ln\frac{5}{4}}$$

16: min/max $f = 81x^2+y^2$ if $9 = 4x^2+y^2$

$$\nabla f = \lambda \nabla(4x^2+y^2) = 72x^2+9$$

Notice $f = 77x + 9y$,

and $g := 4x^2 + y^2 = 9$ is fixed

$\nabla f = \langle 154x, 0 \rangle = \lambda \nabla g = 2\lambda \langle 4x, y \rangle$

$\Rightarrow \begin{cases} 154x = 8\lambda x \\ 0 = 2\lambda y \end{cases}$, so $y = 0$ or $\lambda = 0$

$\lambda = 0$: $154x = 8\lambda x = 0 \Rightarrow x = 0 \Rightarrow f = 9$

$y = 0$: $9 = 4x^2 + y^2 = 4x^2 \Rightarrow x^2 = \frac{9}{4}$

$\Rightarrow f = 9 + 77x^2 = 9 + 77 \cdot \frac{9}{4} =$

$9 + \frac{693}{4} = \frac{693 + 36}{4} = \frac{729}{4}$.

$$\begin{array}{r} 6 \\ 77 \\ \cdot\ 9 \\ \hline 693 \end{array}$$

Comparing the values, we get

minimum is 9 & maximum is $\frac{729}{4}$